

Midterm Project Report: Time Series Analysis of Column H Data

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GitHub Repository - [Click Here](#)

1. Introduction

In this project, I analyzed **Column H (column number 7)** from the "arma_dataset_mid_term.csv" file. The dataset contains daily data from January 1, 2026 to December 31, 2026, with 365 observations total. The main goal of this project was to check the data, find the best mathematical model, and use that model to predict the next 9 future data points. I used ARMA and ARIMA forecasting methods for this analysis.

2. Data Overview

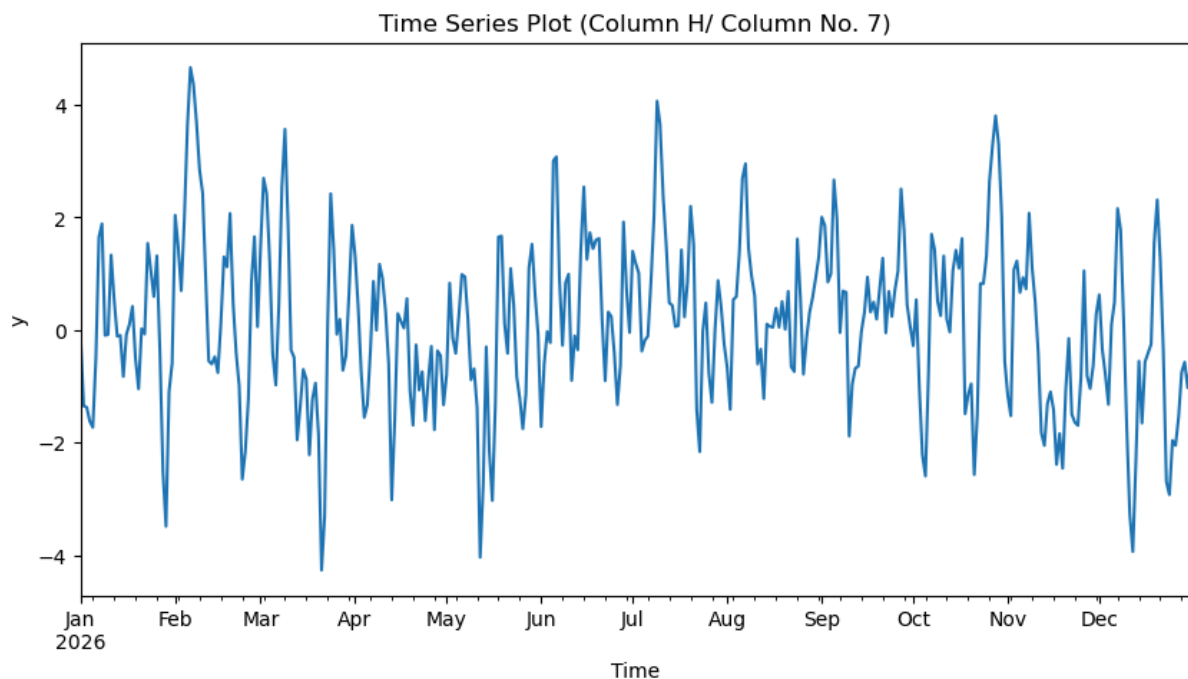
I loaded the dataset and selected Column H as my time series. Here's what the first few values look like:

```
--- My Dataset Head ---
2026-01-01    -0.138456
2026-01-02    -1.348909
2026-01-03    -1.373197
2026-01-04    -1.620728
2026-01-05    -1.731096
Name: 7, dtype: float64

--- My Dataset Length ---
365
```

3. Initial Data Check and Stationarity

I created a Time Series Plot with "time" on the x-axis and "y" on the y-axis. Looking at the graph, the data mostly bounced up and down around a central line without going far way up or far way down over time.



To formally check if the data is stationary, I used **the KPSS test**. The KPSS test checks if a time series is stationary around a constant mean.

Result of the KPSS Test are:

- KPSS Statistic: 0.15808021366600225
- **p-value: 0.1**
- Lags: 20
- Critical Values: {'10%': 0.347, '5%': 0.463, '2.5%': 0.574, '1%': 0.739}

Since **the p-value (0.1) is greater than 0.05**, we cannot reject the null hypothesis that the series is stationary. This means our data is **already stationary** and we don't need to do any differencing. So we can **use d=0** in our ARIMA model.

4. Choosing the Best Model

Next, I created the ACF (Autocorrelation Function) and PACF (Partial Autocorrelation Function) plots. These charts help us see how a data point is related to the points that came before it.

To find the exact best numbers for the model, I tested **different combinations of AR (Auto-Regressive) and MA (Moving Average) orders**. I looked at the **AIC (Akaike Information Criterion) and BIC (Bayesian Information Criterion) scores**. The model with the lowest scores is the best one. Since ARMA(1,1) was chosen by both BIC and HQIC, I selected ARIMA(1,0,1) for my analysis.

```
In [61]: print("Best AIC order:", inf.aic_min_order) # Shows best model by AIC
Best AIC order: (np.int64(3), np.int64(1))

In [62]: print("Best BIC order:", inf.bic_min_order) # Shows best model by BIC
Best BIC order: (np.int64(1), np.int64(1))

In [63]: print("Best HQIC order:", inf.hqic_min_order) # Shows best model by HQIC
Best HQIC order: (np.int64(1), np.int64(1))
```

5. Checking the Residuals

After fitting the model, I checked the residuals to make sure my model is good.

This function (**normaltest()**) tests the null hypothesis that a sample comes from a normal distribution. The **p-value is 0.9683**, which is much greater than 0.05. $p\text{-value} > 0.05$ means we fail to reject the null hypothesis. The data **follow a normal distribution**, indicating that my **residuals are normally distributed**. This shows that my model is good.

I also plot the ACF and PACF of the residuals. I plot the ACF and PACF graphs again, but this time just for the residuals. **The graphs showed no clear patterns**, showing that our model is working well.

The Ljung-Box test checks if there's any pattern left in the residuals. We want the p-values to be high (above 0.05). The plot shows that almost all p-values exceed 0.05, indicating that our residuals **appear to be white noise (no patterns left, no autocorrelation)**.

These facts show that our **ARIMA(1,0,1) model** is appropriate for the data.

6. Forecasting Future Points

The original dataset had 365 points (indexed from 0 to 364). I used my model to predict 9 days beyond my data. The prediction and the y values graph show that our ARIMA(1,0,1) model does a great job of forecasting Column 7 (Column H).

7. Model Equation

The ARIMA(1,0,1) model can be written as:

$$y_t = 0.1010 + 0.4690y_{t-1} + \epsilon_t + 0.4980\epsilon_{t-1}$$

8. Conclusion

Based on my analysis, I found that the Column H data is already stationary, so no differencing was needed. The **ARIMA(1,0,1) model** was the best choice **based on BIC**. The model passed all residual checks and successfully predicted the final 9 points. The residuals look like **random noise with no patterns**. Overall, the ARIMA(1,0,1) model successfully captures the patterns in Column H data and can be used for short-term forecasting.